

Causal Inference: Time Series

Project

Task Description: The goal of this task is to learn the causal structure among a set of financial institutions in the S&P 500 (see Table 1 for the list of companies) by observing their prices over the period from 01.03.2010 to 01.03.2026.

Data: Write a code to obtain daily S&P 500 index data from any suitable source, for example the Yahoo Finance historical dataset for the S&P 500 index (GSPC). From this data, you compute log-prices and apply standard preprocessing steps, such as centering and normalizing, and removing missing values.

Methodology: We aim to apply both linear and nonlinear methods to infer causal graphs and compare their performance. This allows us to assess whether linearity is a reasonable assumption for financial data. First, you should assume that the SEM is linear and estimate the causal graph using a linear method, such as Granger causality. Next, you assume a nonlinear SEM and estimate the graph using directed information. The choice of specific algorithms is left to you.

Time-varying Structure: Since the observation window is long, it is natural to expect that the causal structure evolves over time. Therefore, you should divide the dataset into four subperiods of 4 years each and estimate the causal graph separately for each window. This allows us to analyze how the graph changes across time.

Sample Complexity Reduction: As discussed in the lectures, estimating directed information typically requires large sample sizes. To reduce sample complexity, you can assume that the process is Markov of order $r \in \{1, 2, 3\}$. Furthermore, you can reduce the conditioning set by applying Lemma 1.1 from the lecture notes and assuming that the maximum number of parents per node is $K \in \{4, 5\}$. Motivated by the Black–Scholes model, you may assume that log-prices are jointly Gaussian. You may try other distribution models.

Estimation: For estimation purposes, independent samples are required. Over a given period (e.g., 2010–2014), daily returns provide approximately 1260 observations. One idea to have such independent data is to assume different weeks (5 market days) are independent that is $1260/5 \approx 250$ independent data points for estimation.

Tree Approximation: Finally, you compute the best tree-approximation of the learned causal graph for the last period (01.03.2022 to 01.03.2026).

Statistical Testing and Sensitivity Analysis: To determine whether an estimated causal link is non-zero, you may use either statistical hypothesis testing or a thresholding approach. In either case, a sensitivity analysis is required by varying the threshold (or p-value) and recording the resulting changes in the inferred graph.

Visualization: For presentation, you plot the estimated graphs as well as their edge densities across the four time periods. you further analyze densities within and between sectors over time. For sensitivity analysis, you plot densities as a function of different thresholds or p-values for a

fixed period, Markov order, and maximum number of parents.

Report: Finally, you are required to submit both your code and a written report in PDF format as a single zip file. In the report, you should explain the methodology used and present and discuss the results.

Table 1: High-Liquidity S&P 500 Companies by Sector

Sector	Ticker	Company
Communication Services	GOOGL	Alphabet Inc. Class A
Communication Services	META	Meta Platforms Inc.
Communication Services	NFLX	Netflix Inc.
Communication Services	DIS	Walt Disney Company
Communication Services	CMCSA	Comcast Corporation
Communication Services	TMUS	T-Mobile US Inc.
Energy	XOM	Exxon Mobil Corporation
Energy	CVX	Chevron Corporation
Energy	COP	ConocoPhillips
Energy	SLB	SLB (Schlumberger)
Energy	EOG	EOG Resources Inc.
Energy	MPC	Marathon Petroleum Corporation
Financials	JPM	JPMorgan Chase & Co.
Financials	BAC	Bank of America Corporation
Financials	MS	Morgan Stanley
Financials	GS	Goldman Sachs Group Inc.
Financials	C	Citigroup Inc.
Financials	WFC	Wells Fargo & Company
Information Technology	AAPL	Apple Inc.
Information Technology	MSFT	Microsoft Corporation
Information Technology	NVDA	NVIDIA Corporation
Information Technology	AVGO	Broadcom Inc.
Information Technology	AMD	Advanced Micro Devices Inc.
Information Technology	ORCL	Oracle Corporation
Health Care	UNH	UnitedHealth Group Inc.
Health Care	JNJ	Johnson & Johnson
Health Care	LLY	Eli Lilly and Company
Health Care	ABBV	AbbVie Inc.
Health Care	PFE	Pfizer Inc.
Health Care	MRK	Merck & Co. Inc.